

Guller Yazgi Akata

Applicant

Undergraduate Student Researcher, The State University of New York at Buffalo — Buffalo, NY
gullerya@buffalo.edu — +1-(201)-895-8784 — www.linkedin.com/in/yazgiakata

RESEARCH INTERESTS

As a prospective PhD student, I am broadly interested in algorithmic fairness, and more specifically promoting constructive cross-community interactions and reducing polarization through digital ecosystems.

EDUCATION & SCHOLARSHIPS

The State University of New York at Buffalo , Buffalo, NY	August, 2022 — May, 2026
Bachelor of Science in Computer Science — Minor in Philosophy, Politics and Economics	Cumulative GPA: 3.7/4.00
Yildizlar Education Fund Scholar	
Darussafaka Schools , Istanbul, Turkiye	September 2014 — June 2022
Valedictorian	

SCHOLARSHIPS & AWARDS

Yildizlar Education Fund Scholar (2022-2026)	Buffalo, NY
Scholarship to SUNY at Buffalo covering full tuition, board, flights and stipend provided to gifted, but underprivileged students.	
Darussafaka Schools Student Scholarship (2014-2022)	Istanbul, Turkiye
Full scholarship to Turkiye's first educational NGO, which provides quality education and boarding opportunity to talented but financially disadvantaged kids with deceased fathers and/or mothers. Darussafaka had 0.06% acceptance rate in 2014.	
Kennedy-Lugar Youth Exchange and Study Program (2019-2020)	NJ, USA
Funded by the United States Department of State, scholarship offered to students from countries of strategic importance to live and study in the US for an academic year to promote mutual understanding among cultures.	
FIRST Robotics Competition World Championship Engineering Inspiration Award (2019)	Houston, TX
First team from Turkiye to be globally recognized at the world championship for demonstrating outstanding success in advancing respect and appreciation for engineering within a team's school or organization and community.	2019

ACADEMIC EXPERIENCES

Department of Computer Science & Engineering, SUNY at Buffalo	Buffalo, NY
<i>Undergraduate Student Researcher advised by Dr. Atri Rudra</i>	January 2025 — Present
Conducted research under Dr. Atri Rudra on the reliability of a linear algebraic operator simulating gated-convolution architectures. Proposed a parity-based error-correction method that adds auxiliary data into the input and model parameter, and recovers corrupted outputs without recomputation, or restructuring the operator. I am currently formalizing the approach.	
<i>Head Undergraduate Teaching Assistant</i> <i>CSE116: Introduction to Computer Science II</i>	May 2024 — January 2026
Conducted the leveraging of a tool used for streamlining the process of an assessment used to assess technical communication skills of students. Assembled questions targeting students' understanding, and oversaw the grading of these assessments effectively communicating with both instructors and teaching assistants. Completed administrative tasks.	
<i>Undergraduate Teaching Assistant</i> <i>CSE486: Distributed Systems</i>	January 2026 — Present
<i>CSE220: Systems Programming</i>	August 2024 — Present
<i>CSE116: Introduction to Computer Science II</i>	January 2024 — January 2026
<i>CSE115: Introduction to Computer Science I</i>	August 2023 — December 2023
Addressed concepts such as compiler and linker invocation, memory management, threads, synchronization primitives, data structures, classes, and graph traversal algorithms by leading several weekly lab sections and office hours covering weekly lab assignments. Collaborated with 30+ course staff to ensure smooth operation of the course, contributed to the improvement of each course by consistently providing feedback during the weekly TA meetings, and proctored and graded assessments while ensuring academic integrity and fair opportunity.	

SKILLS

- **Programming:** Python, C, Go, Java, JavaScript
- **Software:** Emacs, Visual Studio Code, IntelliJ Idea
- **Frameworks:** React, Django

OTHER EXPERIENCES & VOLUNTEERISM

UBHacking

Advisor

Director

Sponsorship Lead

Buffalo, NY

March 2025 — Current

March 2024 — March 2025

August 2023 — March 2024

- Directed UB Hacking, a 500+ participant weekend-long hackathon focused on computing for social good, coordinated workshops and judging with faculty, trained and managed an 80% new organizing team, oversaw fundraising and budgeting to ensure sustainability, advised subsequent organizers, and led sponsor communications and onboarding.

Campus Living at SUNY at Buffalo

Buffalo, NY

Community Assistant at Hadley & South Lake Villages

August 2024 — August 2025

- Fostered an inclusive and supportive living environment for 69 residents by organizing community-building events, providing day-to-day guidance, and addressing resident concerns with empathy and care.

International Student Advisory Council at SUNY at Buffalo

Buffalo, NY

Founding Board Member

October 2022 — Present

- Advocate for the international student body at UB, serve as an accessible resource to students, provide guidance, assistance and feedback to the International Students Services office on student inclusion.

Young Guru Academy

Hatay, Turkiye

Volunteer for World Science Movement

January 2025

- Led workshops to encourage a love for science for underprivileged elementary school students in the still recovering areas from the earthquake of February 6th, 2023 in Hatay, Turkiye.

AFS intercultural Programs

Washington D.C.

Workshop Facilitator & Volunteer

2023—2025

- Facilitated workshops on cultural transition and intercultural communication for the incoming class of youth ambassadors from various countries.

CLASS PROJECTS

Kademlia

Buffalo, NY

- Inspired by Kademlia, implemented a peer-to-peer distributed hash table (DHT) in Go, utilizing XOR based node lookup for routing table maintenance.

CSE Interview Help Tool

Buffalo, NY

- Cleaned, processed and analyzed approximately 25,000 lines of student interview data using data analysis tools and TF-IDF method in Python. Trained artificial intelligence driven tool using KNN classification to help students improve their technical communication skills and success in assessments.

Spinderfy

Buffalo, NY

- Designed and developed both the front and back end of a web application that allows user to like and skip music, recommended based on user's past behavior, using Figma, React, Vite and PHP. Integrated Spotify and Gemini API into the process.

REFERENCES

Dr. Atri Rudra

Katherine Johnson Chair in Artificial Intelligence, Department of AI and Society at SUNY at Buffalo, Buffalo, NY

E-mail: atri@buffalo.edu

Dr. Ethan Blanton

Assistant Professor of Teaching, Department of Computer Science and Engineering at SUNY at Buffalo, Buffalo, NY

E-mail: eblanton@buffalo.edu

Dr. Paul Dickson

Assistant Professor of Teaching, Department of Computer Science and Engineering at SUNY at Buffalo, Buffalo, NY

E-mail: pauldick@buffalo.edu

Alan Hunt

Professor of Practice, Department of Computer Science and Engineering at SUNY at Buffalo, Buffalo, NY

E-mail: ahunt@buffalo.edu